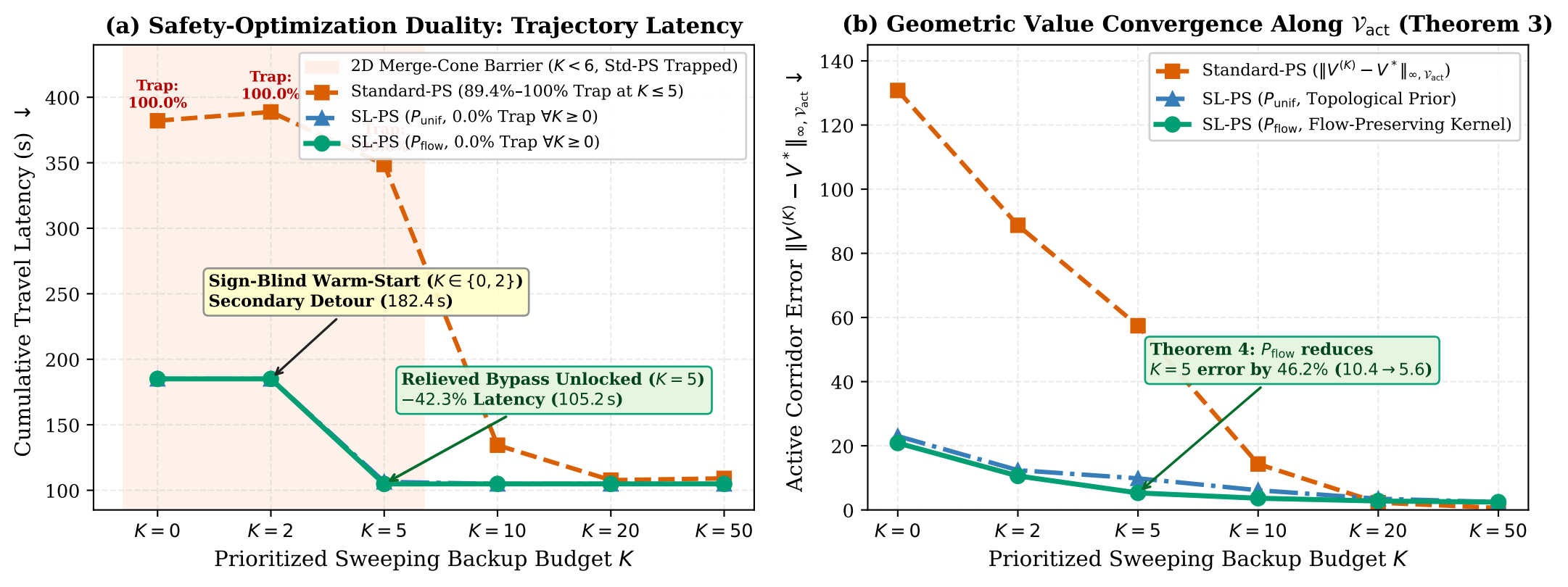

(a) Safety-Optimization Duality: Trajectory Latency
2D Merge-Cone Barrier ($K < 6$, Std-PS Trapped)
Standard-PS (89.4%–100% Trap at $K \leq 5$)
SL-PS ($P_{\text{unif}}$, 0.0% Trap $\forall K \geq 0$)
SL-PS ($P_{\text{flow}}$, 0.0% Trap $\forall K \geq 0$)
Trap: 100.0%
Trap: 100.0%
Sign-Blind Warm-Start ($K \in \{0, 2\}$) Secondary Detour (182.4 s)
Relieved Bypass Unlocked ($K = 5$) −42.3% Latency (105.2 s)
Cumulative Travel Latency (s) →
Prioritized Sweeping Backup Budget $K$
(b) Geometric Value Convergence Along $\mathcal{V}_{\text{act}}$ (Theorem 3)
Standard-PS ($\|V^{(K)} - V^*\|_{\infty, \mathcal{V}_{\text{act}}}$)
SL-PS ($P_{\text{unif}}$, Topological Prior)
SL-PS ($P_{\text{flow}}$, Flow-Preserving Kernel)
Theorem 4: $P_{\text{flow}}$ reduces $K = 5$ error by 46.2% (10.4 → 5.6)
Active Corridor Error $\|V^{(K)} - V^*\|_{\infty, \mathcal{V}_{\text{act}}}$ →
Prioritized Sweeping Backup Budget $K$